
Laboratory Guide

Lesson 03 — LPWAN Technology Selection

Communication Technologies for the Internet of Things

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Catarina Silva

ESTGA, University of Aveiro

Estimated duration: 2 hours

Contents

Context	2
Learning Outcomes	2
System Architecture	2
1 Technology Reference Table	3
2 Building the Decision Flow	4
3 Application to Scenarios	6
Questions for Discussion	7
References	7

Context

The learning outcome associated with syllabus item C3 is to *select and apply the LPWAN communication standards best suited to the requirements and constraints of a specific solution*. This session exercises that verb directly: an LPWAN network is not deployed, rather a decision is made as to which one to use.

The session uses no equipment. The technologies covered in the lecture require operator infrastructure or dedicated gateways, which are not available in the laboratory. The problem an engineer faces at the design stage is, however, not one of assembly but of comparison and decision, and that is fully reproducible in a simulation environment.

All logic is built in Node-RED by connecting blocks. No code is written.

Required material

- Personal computer with Node-RED installed and the `node-red-dashboard` package (see Lesson 02, prior preparation section).
- No laboratory equipment is used.

Learning Outcomes

By the end of the session, the student should be able to:

- Translate application requirements into technical parameters comparable across LPWAN technologies;
- Justify the exclusion of a technology on the basis of an unmet requirement;
- Recognise the trade-offs between data rate, range, battery life and cost, and identify that no technology optimises them simultaneously;
- Build in Node-RED a decision flow comprising data entry, rules and result presentation.

System Architecture

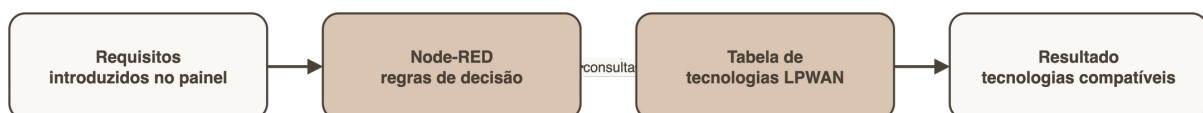


Figure 1: Architecture of the system to be built

1. Technology Reference Table

20 min

Objective

Consolidate, in a single table, the parameters of the technologies covered in the lecture, so as to make them comparable.

Complete the table below from the lecture slides. Missing values should be taken from the slides corresponding to each technology.

Technology	Data rate	Range	Spectrum	Main constraint
LoRaWAN	up to 50 kbit/s	up to 15 km	Unlicensed	Regulatory duty cycle
Sigfox	≈ 100 bit/s	up to 50 km	Unlicensed	140 messages/day, 12 bytes
Weightless-N	low	> 5 km	Unlicensed	Unidirectional
Weightless-P	low	> 2 km	Unlicensed	Shorter battery life
RPMA	low	high	Unlicensed (2.4 GHz)	Restricted ecosystem
EnOcean	very low	short	Unlicensed	Energy harvested from environment
Z-Wave	9–40 kbit/s	≈ 50 m	Unlicensed	Home-automation range
LTE-M	up to 1 Mbit/s	≈ 11 km	Licensed	Per-device cost
NB-IoT	≈ 150 kbit/s	≈ 11 km	Licensed	High latency

Reading note

The figures given correspond to favourable propagation conditions and are intended for comparing orders of magnitude. In a real deployment, effective range depends on terrain, antenna height and spectrum occupancy.

2. Building the Decision Flow

40 min

Objective

Build in Node-RED a flow that takes the requirements of an application and presents the technologies that satisfy them.

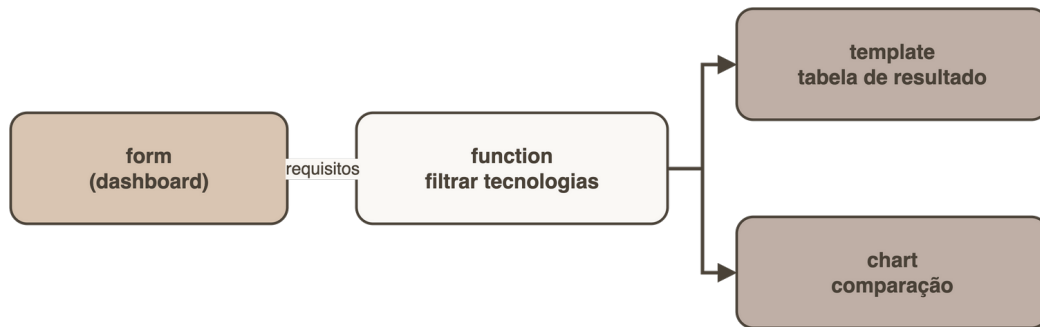


Figure 2: Node-RED flow to be built

Procedure

1. Drag a **form** node from the *dashboard* section. Define four numeric fields: **debito** (kbit/s), **alcance** (km), **autonomia** (years) and **mensagens** (per day).
2. Drag a **function** node and connect it to the output of the **form**. The code is provided and is only to be pasted:

```

var r = msg.payload;
var tec = [
  {nome:"LoRaWAN", debito:50, alcance:15, autonomia:10, msgs:100},
  {nome:"Sigfox", debito:0.1, alcance:50, autonomia:10, msgs:140},
  {nome:"RPMA", debito:8, alcance:20, autonomia:10, msgs:1000},
  {nome:"Z-Wave", debito:40, alcance:0.05, autonomia:5, msgs:10000},
  {nome:"LTE-M", debito:1000, alcance:11, autonomia:5, msgs:100000},
  {nome:"NB-IoT", debito:150, alcance:11, autonomia:10, msgs:100000}
];
msg.payload = tec.filter(function(t) {
  return t.debito >= Number(r.debito)
    && t.alcance >= Number(r.alcance)
    && t.autonomia >= Number(r.autonomia)
    && t.msgs >= Number(r.mensagens);
});
return msg;

```

3. Drag a **template** node from the *dashboard* section to display the list of compatible technologies.
4. Drag a **chart** node to compare graphically the data rate of the selected technologies.
5. Press **Deploy** and open <http://127.0.0.1:1880/ui>.

On the provided code

The block above implements filtering by minimum requirements. It is provided in full and is not itself assessed: what is assessed is the interpretation of the results it produces. The

values in the `tec` array may be adjusted as the table in Section 1 is completed.

3. Application to Scenarios

40 min

Objective

Apply the flow to concrete scenarios and justify each decision on technical grounds.

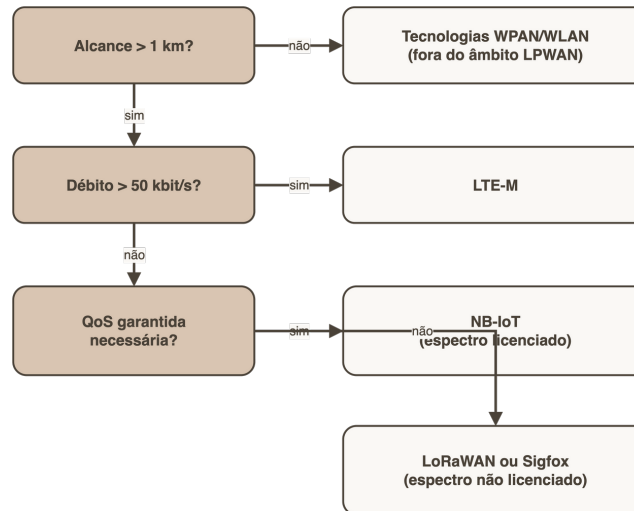


Figure 3: Reference decision tree

For each scenario, enter the requirements in the panel, record the technologies returned and write a two to three line justification for the technology chosen.

No.	Scenario
1	Urban water metering. Thirty thousand meters across a municipality. One daily reading of 8 bytes per meter. Ten years of battery life required without replacement. Per-device cost is decisive.
2	Vineyard monitoring. Two hundred soil moisture and temperature sensors spread over 40 hectares, 6 km from the estate headquarters. One 20-byte reading every 15 minutes. No reliable cellular coverage on site.
3	Port container tracking. Five hundred containers moving within a 5 km area. Position reported every 5 minutes. Message delivery confirmation and guaranteed quality of service required.
4	Lighting switches in an office building. One hundred and twenty switches. Occasional actuation, with no periodicity. Requirement of no battery replacement during the building life cycle.
5	Gas leak detection in an industrial network. Eighty sensors within a 2 km radius. An alarm must reach the operator in under 10 seconds.

Guidance

In at least one scenario the flow will return more than one compatible technology, and in another it may return none. Both cases are intentional. Where more than one admissible solution exists, the tie-breaker is cost and ecosystem maturity; where none exists, the requirement to be renegotiated with the client should be identified.

Further work

Add to the flow a node that, when no technology satisfies the requirements, identifies which requirement is responsible for excluding the largest number of candidates.

Questions for Discussion

1. Scenario 4 admits a battery-free solution. Which, and at what cost in range?
2. Why is Sigfox excluded from scenario 3, even though its range is sufficient?
3. Under what conditions is the higher cost of a licensed-spectrum solution justified?
4. What would happen to scenario 2 if periodicity changed from 15 minutes to 1 minute?

References

- Gonçalves, P. *Apresent3 — LowPowerWAN*, TCIoT, previous editions.
- Adelantado, F. *et al.* (2017). Understanding the Limits of LoRaWAN. *IEEE Communications Magazine*, 55(9). Available as [Leituras/1607.08011.pdf](#).
- Node-RED documentation: <https://nodered.org/docs/>
- Dashboard nodes: <https://flows.nodered.org/node/node-red-dashboard>